AcademAI: An Automated Essay Evaluation System Using a Transformer-Based Model

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Abstract*—The increasing use of AI-assisted writing tools has created challenges for educators in maintaining academic integrity and ensuring fair and consistent essay evaluation. Existing Automated Essay Evaluation (AEE) systems partially address these issues but often lack integrated mechanisms to detect AI-generated content, leading to unreliable assessments. This study developed AcademAI, a web-based platform that uses Google's Gemini transformer-based model to generate rubric-aligned scores and detect AI-generated text. The research aimed to create an AI-driven scoring mechanism based on faculty rubrics, integrate deep learning techniques to differentiate human-written and AI-generated essays, and measure the agreement between human raters and AI scoring using statistical analyses. A mixed-method approach was used, involving dataset preprocessing, model training on 29,145 text samples, survey evaluation, and computation of Cohen's Kappa, RMSE, and paired t-tests. Results showed strong model performance, with precision, recall, and F1-scores reaching 83–85%, supported by consistent classification behavior. Agreement analysis revealed substantial to perfect alignment between professors and AI scores, with Cohen's Kappa values ranging from 0.231 to 1.0. RMSE values between 0.32 and 0.59 indicated close scoring proximity, while all paired t-tests showed p-values above 0.05, confirming no significant scoring differences. Overall, AcademAI demonstrated reliable performance and potential as a supplementary academic assessment tool.

Keywords—*Academic Integrity, AI Detection, Automated Essay Evaluation, Deep Learning, Transformer-Based Model*

I. INTRODUCTION

In education, the assessment of student learning played a crucial role. It helped educators evaluate how well students understood the material taught in class. One way they did this was through essay exams, which were an effective tool for assessing academic achievement, integration of ideas, and the ability to recall. Educators appreciated essays for showcasing higher-order thinking skills such as reasoning, creativity, and analysis. According to Bane Raman Raghunath et al. (2022) [1], essays support the demonstration of complex cognitive processes that cannot be captured through objective tests alone. However, with the rapid and continual advancement of technology, particularly in artificial intelligence (AI) and natural language processing, the educational landscape has experienced significant shifts that influence how students approach academic writing and how educators assess such outputs.

While technology has enhanced educational access and efficiency, it has also enabled practices that threaten academic integrity. Students now have the ability to generate written content using AI tools that can produce grammatically correct and well-structured essays within seconds. This ease of content generation challenges the reliability of essay assessments as authentic measures of student learning. Educators increasingly struggle to differentiate between genuinely written essays and those generated by AI, compromising fairness in grading and potentially diminishing students' writing proficiency.

Recent studies highlight the severity of this issue. Surveys in the United Kingdom and Sweden found that 67% of secondary students and 35% of university students regularly used AI for academic tasks [2]. These trends reveal the growing difficulty of maintaining academic honesty and ensuring consistent, objective evaluation of student essays. Manual grading is also time-consuming and prone to human error, making the assessment process even more challenging.

Several studies have explored automated essay scoring (AES) systems as potential solutions. Early models such as Project Essay Grade (PEG) by Page [3] relied on surface-level linguistic features, while later systems like e-rater incorporated natural language processing for better accuracy [4]. IntelliMetric introduced neural networks but showed inconsistencies when scoring highly creative responses [5]. Shermis and Burstein [6] emphasized that while AES systems can approximate human scoring, they remain vulnerable to formulaic writing. More recent technologies, including GPT-4 and Google's Gemini, offer advanced comprehension and semantic analysis [7], [8], yet AI-detection tools remain unreliable, often producing false positives or overlooking sophisticated AI-generated texts [9]. Consequently, there remains a need for a system that integrates both accurate essay scoring and reliable AI-generated content detection.This demonstrated how prominent academic dishonesty and misconduct had become in the academic environment due to the advancement of technology. Although it made students' lives easier, it also made them less competent when it came to writing. This was because they had normalized the thought of "I will just get an idea, but I won't copy it all" or submitted works generated by AI out of laziness, rather than taking the time to read and comprehend the materials given and apply what was taught in class.

To address these challenges, the present study proposes the development of a web-based automated essay scoring and AI-detection system using Google's Gemini. The system aims to evaluate essays based on academic writing criteria while simultaneously detecting the likelihood of AI-generated content. Through this integrated approach, the project seeks to provide educators with an alternative, reliable, efficient, and objective tool for evaluating student essays, thereby enhancing academic integrity, promoting authentic student learning, and improving the overall assessment process.

II. Related Studies

The evaluation of student essays has undergone a significant evolution from manual grading to the adoption of automated systems. Traditionally, essay assessment relied on human evaluators, which, while effective, was time-consuming, inconsistent, and often influenced by subjective bias. Early automated approaches, such as Project Essay Grade (PEG) introduced by Page [10], used shallow linguistic features like word frequency and sentence length to generate scores. These systems provided efficiency but lacked semantic understanding and context awareness, limiting their ability to assess the deeper quality of student writing [11].

Recent advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP) have led to more sophisticated Automated Essay Evaluation (AEE) systems. Transformer-based language models such as BERT, GPT, and Google Gemini employ self-attention mechanisms to capture context, structure, and meaning within text, enabling more human-like evaluation of essays [11]. These innovations allow AEE systems to provide scalable, efficient, and consistent scoring aligned with faculty-defined rubrics, representing a significant improvement over earlier statistical or rule-based models. Rubric-based scoring remains central to ensuring transparency and fairness, with studies emphasizing that analytic rubrics allow detailed assessment of multiple dimensions such as content, organization, mechanics, and style [12]. However, many existing systems either rely on fixed rubrics or offer limited customization, which constrains their adaptability across different courses or institutions.

Another important area in recent research involves distinguishing AI-generated text from human-written essays. Studies by Georgiou [13] demonstrated that linguistic patterns and measurable differences exist between AI and human writing, which can be detected through a combination of statistical modeling and deep learning techniques. Despite this, modern AI tools are increasingly capable of mimicking human writing, especially with small or limited datasets, making accurate detection a persistent challenge. This highlights the need for integrating deep learning-based detection techniques with scoring systems to ensure academic integrity.

Despite the progress, limitations remain in existing AEE platforms. Established systems like ETS's e-rater provide automated scoring but often operate as "black boxes," limiting transparency and lacking robust AI-authorship detection features. Furthermore, most studies focus either on scoring or AI-detection separately, leaving a gap in research for systems that combine both functionalities in a web-based, user-friendly platform.

III. Methodology

This section outlines the methodological framework employed in the development, implementation, and evaluation of the proposed web-based automated essay scoring and AI-content detection system. The methodology integrates data collection procedures, model training techniques, evaluation tools needed, and system evaluation strategies necessary to ensure accuracy, reliability, and usability. It describes the system architecture, the application of Google's Gemini for rubric-based scoring, the deep learning approach used to distinguish AI-generated from human-written text, and the procedures for determining inter-rater agreement through Cohen's Kappa coefficient. The methods presented provide the foundation for validating the system's performance and establishing its effectiveness in supporting fair and objective academic assessment.

A. Statistical Tool

To evaluate the reliability of scores assessed by A.I. and humans, Cohen's Kappa was used. This metric measures inter-rater agreement while accounting for chance, offering a more accurate view than simple percentage match. In essay evaluation, where scores are grouped into categories or bands, Cohen's Kappa helps determine how well the AI aligns with human judgment. Higher values indicate stronger agreement, highlighting the system's accuracy and guiding further improvements where needed.

$$k = \frac{P(A) - P(E)}{1 - P(E)}$$

B. Data Analysis

In the data preparation, the dataset used for this study is the *Training_Essay_Data.csv taken from Kaggle.com*, which consists of labeled text samples categorized as either AI-generated or human-written. This comprehensive dataset comprises a total of 29145 unique values (texts) with 3 columns, 11637 rows for AI-generated essays (1) and 17508 rows for human-written essays (0).

Preprocessing steps were applied to ensure data consistency, including converting text to lowercase to maintain uniformity, removing special characters and extra spaces to eliminate noise and improve model efficiency. After preprocessing, the dataset was divided into training and testing subsets using an 80-20 split. Stratified sampling was employed to ensure that both AI-generated and human-written texts were proportionally represented in both subsets. This approach helped prevent data imbalance and ensured a fair evaluation of model performance.

To prepare the textual data for deep learning, tokenization was performed using the Keras Tokenizer, with the vocabulary size limited to the 20,000 most frequently occurring words. Each text sample was converted into numerical sequences corresponding to word indices. To ensure uniform input for the neural network, all sequences were padded or truncated to a maximum length of 500 tokens, preventing variations in text length from affecting model performance.

The AI–human text detection model processes tokenized text sequences of fixed length 500 and employs an embedding layer (vocabulary size 20,000, embedding dimension 128) to convert words into dense vectors. Contextual dependencies are captured using a bidirectional LSTM layer with 64 units, followed by a dropout layer (rate 0.5) to reduce overfitting. A fully connected dense layer with a sigmoid activation function performs binary classification to determine whether a text is AI-generated or human-written.

The AI–human text detection system was trained using a neural network framework that learns linguistic patterns from the tokenized dataset to distinguish between human-written and AI-generated content. Following data preparation and text tokenization, the textual data—represented as numerical sequences with a consistent length of 500 tokens—served as the input for model training. The dataset, divided into training and testing subsets with stratified sampling to preserve proportional representation of AI-generated and human-written essays, ensured balanced learning and fair performance evaluation.

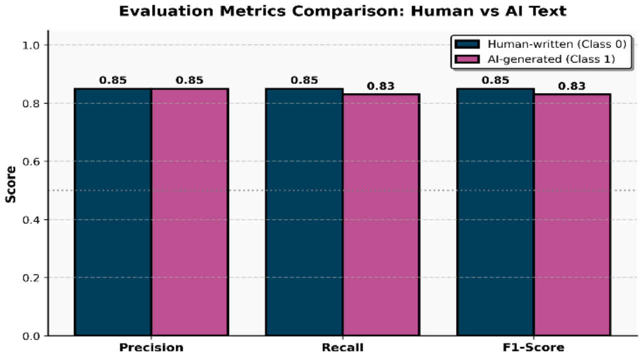
The training framework combined rule-based pattern matching, heuristic scoring, and statistical text analysis to extract meaningful features from the tokenized sequences. Rule-based pattern matching identified deterministic textual markers frequently associated with AI-generated content, such as repetitive syntactic structures, uniform sentence lengths, and characteristic phrase patterns. These rules were derived from iterative examination of the tokenized training samples, providing an interpretable mechanism for detecting stylistic regularities.

In parallel, heuristic scoring was applied to the tokenized sequences to evaluate linguistic attributes, including lexical diversity, sentence variability, semantic coherence, and redundancy. Each attribute was weighted proportionally to reflect its significance in differentiating AI-generated and human-written text, complementing the deterministic rules and providing flexibility for nuanced patterns.

Statistical text analysis further enhanced the system by examining quantitative patterns within the corpus, including word frequency distributions, n-gram uniqueness, perplexity measures, and syntactic dispersion. These analyses leveraged the tokenized representations to detect subtle differences in text structure and style that might not be captured by rule-based or heuristic methods alone.

The model was trained using the Adam optimizer with a learning rate of 0.001, and binary cross-entropy was used as the loss function for binary classification. Training was conducted for 10 epochs with a batch size of 32. During training, the model automatically learned linguistic and contextual features from the dataset, enabling it to distinguish patterns associated with human-written and AI-generated text.

Figure 1 shows the model evaluation results. The model's performance was assessed using the remaining 20% of unseen data, demonstrating strong classification capability for both human-written and AI-generated text. Precision scores were identical at 85%, indicating balanced accuracy in predicting each class. Recall and F1-scores showed a slight advantage toward detecting human-written content, with human samples scoring 85% and AI-generated samples slightly lower at 83%, suggesting that the model misses more AI text than human text



Summary of Evaluation Metrics

Class	Precision	Recall	F1-Score
Human-written (0)	0.85	0.85	0.85
AI-generated (1)	0.85	0.83	0.83

Fig. 1. Model Evaluation

Figure 2 shows the confusion matrix of the model. The confusion matrix further illustrates this trend. Out of 5,829 test samples, 3,285 human-written texts were correctly classified, while 217 were incorrectly flagged as AI-generated. Meanwhile, the model accurately identified 1,471 AI-generated texts but misclassified 856 of them as human-written, indicating that certain AI-generated responses closely mimic human writing.

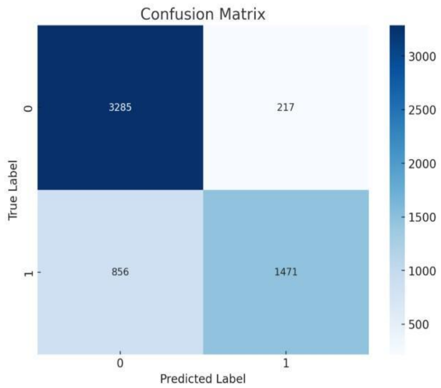


Fig. 2. Confusion Matrix

Overall, the model performs well, though reducing false negatives remains an area for improvement to enhance AI-text detection reliability.

The confusion matrix (Figure 2) indicates that 856 AI-generated essays were initially misclassified as human-written, reflecting a high false-negative rate. To improve AI-detection reliability, a threshold calibration procedure was implemented using AI-probability scores generated by the system. Each essay was assigned a probability representing the likelihood of being AI-generated versus human-written (e.g., 25% AI / 75% Human or 45% AI / 55% Human), with an initial threshold set at 50% AI. Controlled, iterative adjustments of the threshold were conducted in small increments (e.g., 45%, 40%), and resulting changes in recall and precision were evaluated to assess the tradeoff between correctly identifying AI-generated texts and avoiding misclassification of human-written essays. Additionally, refinement of heuristic and statistical feature weights strengthened linguistic signals associated with AI-generated writing. These strategies enhanced detection of borderline AI-generated essays while maintaining overall accuracy. Nevertheless, overlap between AI-generated and human-written linguistic patterns may persist in borderline cases, and threshold calibration mitigates—but does not entirely eliminate—such occurrences.

Specifically addressing the 856 AI-generated essays misclassified as human-written in the confusion matrix, threshold calibration was implemented to reduce the false-negative rate. Borderline essays that produced intermediate AI-probability scores were re-evaluated using adjusted classification thresholds. For example, essays scoring 45% AI / 55% Human were initially labeled as human-written under the 50% threshold. By lowering the threshold to 45%, several borderline cases were correctly reclassified as AI-generated, thereby improving recall for the AI class. Although this adjustment slightly increased false positives, the trade-off was considered acceptable given the objective of minimizing undetected AI-generated submissions. Consequently, the refined threshold improves the system's sensitivity to AI-generated writing patterns and enhances the reliability of AI-detection results.

C. *System Architecture*

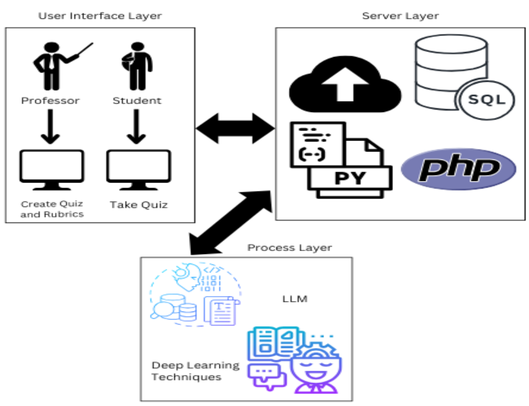


Fig. 3. System Architecture

Figure 3 shows the system architecture of the AcademAI automated essay evaluation system, presenting its core components and workflow. It illustrates how essays are received, processed, and scored. Users first input their personal information, create accounts, and supply essay questions, dynamic rubrics, and written responses. The rubric is fully customizable, allowing users to add, modify, or remove scoring criteria.

Once an essay is submitted, the system activates its evaluation pipeline. Google's Gemini model, accessed through the Gemini API, performs rubric-based scoring using prompt-engineered instructions to ensure accurate and criterion-aligned assessment. In parallel, deep learning methods analyze writing patterns to detect potential AI-generated content. Throughout this process, data from both students and instructors undergo preprocessing—cleaning, tokenization, and structuring—to enable precise scoring, coherence evaluation, and authorship classification. The combined workflow ensures a reliable and automated assessment of essay quality and integrity.

D. *Data Source*

The survey participants were carefully chosen selected based on the necessary knowledge and experience relevant to the study to validate the proposed automated essay evaluation that is capable of automatically generating essay scores, detecting AI-generated answers. Six (6) selected professors from different colleges and their respective departments at Western Mindanao State University were chosen as respondents, along with five (5) students each under their supervision. To ensure high-quality insights from professors in this field, it is reasonable to seek respondents with a combination of academic qualifications, practical teaching experience, and experience in grading academic papers.

IV. Results and Discussion

A. *Assessment and Scoring*

The *AcademAI* uses the *google.genai* library with Google's Gemini 2.0 Flash model to perform automated essay evaluation. It simulates an expert evaluator, applying predefined criteria and weights to generate scores (0–100), brief justifications, and improvement suggestions. An overall weighted score is computed, and results are output in structured JSON format for clear interpretation and tracking of performance.

B. *Evaluation and Scoring Evaluation*

To determine the reliability of the AI-generated scores, the system's outputs were compared with those of six professors across different academic departments. The comparison used Cohen's Kappa (κ), a widely accepted metric for measuring inter-rater agreement beyond chance. Higher κ values reflect stronger agreement between the AI and human raters.

Cohen's Kappa was calculated using the formula:

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

where P_o is the observed agreement and P_e is the expected agreement by chance

TABLE I. Cohen's Kappa Evaluation

Professor	κ Value	Strength of Agreement
Prof. 1	0.615	Substantial
Prof. 2	1.0	Perfect
Prof. 3	0.615	Substantial
Prof. 4	0.583	Moderate
Prof. 5	0.231	Fair
Prof. 6	1.0	Perfect

The results, as outlined in Table 1, show that most professors exhibit substantial to perfect agreement with the AI system, indicating a high degree of alignment between automated and human scoring. Only one professor (Prof. 5) falls under the fair agreement level, suggesting some variation in scoring style. Overall, the strong κ values support the reliability of the AI model as a consistent scoring tool.

C. Root Mean Squared Error (RMSE)

RMSE was calculated to measure numerical differences between the AI and professor scores. Lower RMSE values indicate closer alignment between the two sets of scores. RMSE was calculated using the formula:

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(X_i - Y_i)^2}{n}}$$

where X_i and Y_i represent AI and human scores, respectively

TABLE II. RMSE

Professor	RMSE
Prof. 1	0.35
Prof. 2	0.32
Prof. 3	0.59
Prof. 4	0.41
Prof. 5	0.52
Prof. 6	0.59

The RMSE values (0.32–0.59) show that the differences between human and AI scores are minimal, as outlined in Table 2. This suggests strong numerical consistency, reinforcing the AI's capability to approximate human scoring patterns across diverse departments.

D. AI Tools Benchmark

The AI detection component of *AcademAI* was compared against three benchmark systems: *Copyleaks*, *ZeroGPT*, and *QuillBot*. Although absolute scores varied due to differing detection algorithms, *AcademAI* demonstrated consistent scoring behavior across all test samples. The Mean Absolute Error (MAE) was 37.14, indicating stable internal reliability.

MAE was calculated using the formula:

$$\text{MAE} = \frac{\sum_{i=1}^{n} | X_i - Y_i |}{n}$$

where X_i and Y_i represent the scores from AcademAI and the benchmark system, respectively.

These results highlight the system's potential for refinement and calibration as more training data becomes available.

E. Paired t-Test

A paired t-test was conducted to determine whether significant differences existed between professor and AI scores across all departments. Since each student's essay received two scores, one from the professor and one from the AI, the paired t-test was an appropriate choice.

The t-statistic was calculated using the formula:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

where $\bar{d}$ is the mean of the differences between paired scores, s_d is the standard deviation of the differences, and n is the number of paired observations.

TABLE III. PAIRED T-TEST

Professor	t-statistic	p-value
Prof. 1	0.875	0.431
Prof. 2	0.0	1.0
Prof. 3	1.18	0.305
Prof. 4	1.10	0.333
Prof. 5	0.39	0.717
Prof. 6	0.137	0.898

All p-values exceeded 0.05, confirming no statistically significant difference between professor and AI scores, as outlined in Table 3. This result supports the conclusion that the AI system's scoring is statistically comparable to human evaluation.

F. Evaluation Tool Reliability

The evaluation instrument was assessed using descriptive statistics and percent agreement to determine inter-rater reliability. All raters—including six professors and the AI—uniformly assigned a rating of "1" (Agree) to each item on the evaluation form. This resulted in zero variance, zero standard deviation, and 100% frequency for a single rating category. Percent agreement is calculated as the number of exact agreements divided by the total number of ratings. In this study, six professors each rated twelve items, yielding a total of 72 ratings, all of which were in agreement. This results in a 100% agreement, indicating complete consensus among the raters and the AI, and providing a clear and interpretable measure of perfect reliability for the evaluation instrument.

Overall, the findings validate the AI system as a dependable and accurate tool for academic essay assessment. The results collectively confirm that the AI scoring system aligns strongly with professor evaluations across all metrics. Cohen's Kappa showed substantial to perfect agreement in most departments; RMSE values indicated minimal numerical error; paired t-tests confirmed no significant differences; and reliability testing reflected complete consistency among raters.

V. CONCLUSION

A. Conclusion

The study has successfully developed a web-based system for essay evaluation with a focus on assessing AI-generated and human-written texts. The system uses Google's Gemini 2.0 Flash AI model to evaluate essays based on several criteria, providing clear, structured feedback and scoring. The AI model's performance in essay evaluation demonstrated high accuracy, with scores closely aligning with human evaluators, suggesting that the model can approximate human judgment within the rubric-based evaluation framework.

The AI text detection model using deep learning techniques to identify text patterns was trained on a comprehensive dataset of human-written and AI-generated texts and achieved strong results. The evaluation metrics and confusion matrix revealed that while the model exhibited strong performance, there were still occasional misclassifications, particularly with AI-generated texts closely resembling human writing.

The comprehensive evaluation of the ***AcademAI*** system demonstrates that it is a reliable and consistent performance within the scope of the current validation dataset. The strong alignment between AI-generated and professor-assigned scores, as evidenced by substantial to perfect Cohen's Kappa values, low RMSE ranges, and non-significant paired t-test results, indicates that the system can effectively approximate human judgment across different academic disciplines included in this study.

Furthermore, the inter-rater agreement observed in the reliability analysis underscores the stability and consistency of the evaluation instrument. Benchmarking against existing AI detection tools also showed that ***AcademAI*** delivers stable and dependable performance. Collectively, these findings suggest ***AcademAI*** has strong potential as a supplementary tool to human grading, with strong potential to support educators by providing objective, efficient, and standardized essay evaluation, while further large-scale validation involving broader datasets and more diverse writing samples is recommended to strengthen the system's generalizability.

B. Recommendations

Future work will focus on expanding the system's capabilities and improving its performance. The AI text detection model should be trained on multilingual and more diverse datasets to handle different languages, writing styles, topics, and formats, improving generalization and reducing false negatives. The essay scoring component should also be further fine-tuned and calibrated using a broader collection of professor-graded essays to better align with human scoring patterns across prompts, difficulty levels, and writing styles. Additionally, adaptive scoring techniques can be explored to allow the system to learn from historical grading patterns, ensuring consistency across large volumes of essays. Expanding the validation dataset by involving more professors and a larger, diverse set of student essays will further enhance the system's reliability, robustness, and generalizability, supporting broader institutional adoption.

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